Problem 3

Problem 3

Problem 1 *Assume that the rate of change (in dollars per day) of the price of shares of stock in the WeSaySo Company (with t in days) is modeled by the equation $r(t) = -3t^2 + 30t - 63$ (note that this is technically a* discrete *function, but prices change so often with stocks that modeling this with a continuous function makes sense). Assume also that the price of a share of stock on day 1 (i.e., $t = 1$) is \$51. Answer the following questions:*

(a) *Find the rate of change of price at $t = 5$.*

(b) *Find the price of a share of stock at $t = 5$.*

(c) *How fast is the rate of change of price changing at $t = 5$?*

(d) *How much did the price of a share of stock change in the first 6 days (i.e., on $[1, 6]$)?*

(e) *What was the greatest rate of change of price during the first 6 days (i.e., on $[1, 6]$)?*

(f) *What was the greatest price of a share of stock during the first 6 days (i.e., on $[1, 6]$)?*